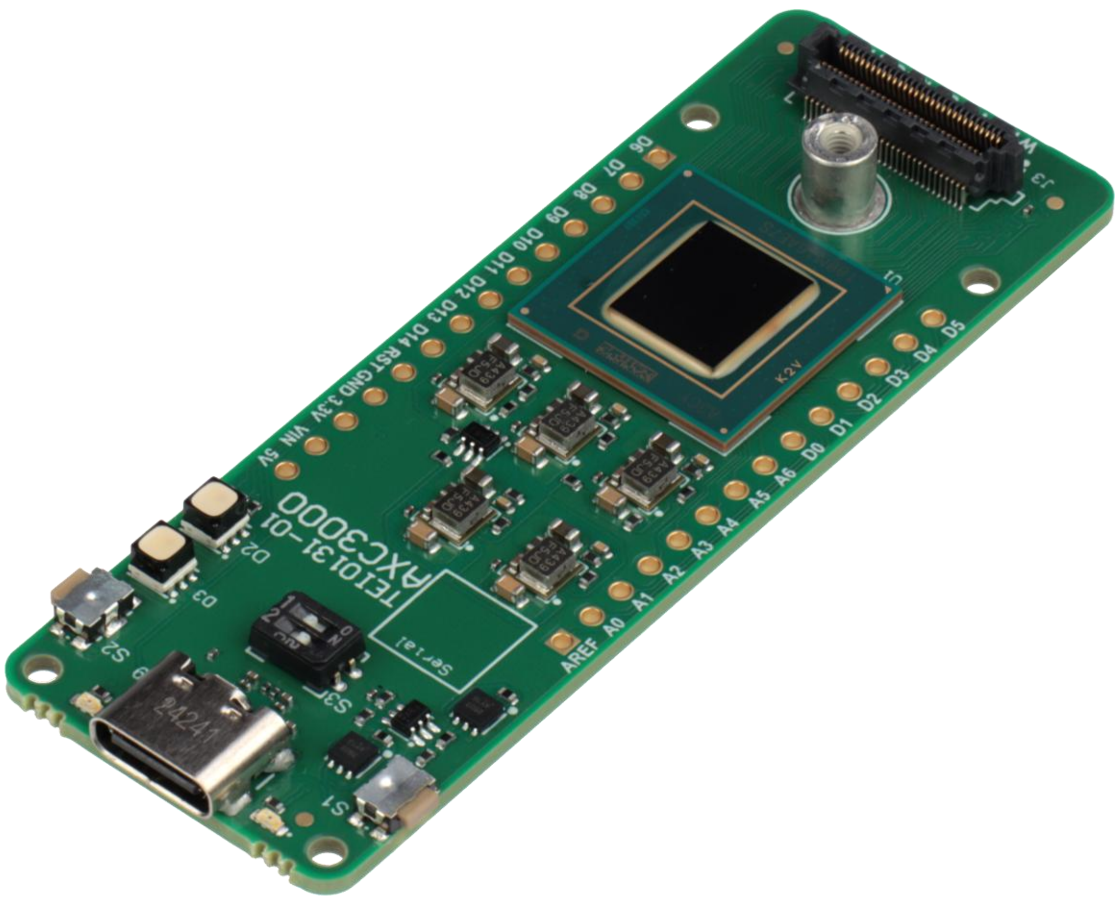
Installation Guide
(North America)



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Prior Version History:	

Please read the legal disclaimer at the end of this document.

Contents

1 Workshop scenarios

There are three different scenarios in which the Agilex 3 labs can be implemented and verified. They are:

CloudLabs – build in the cloud and verify with board farm hardware

Local PC (verification only) – build in the cloud, verify on local hardware

Local PC – build with a local PC and verify with local hardware

All three have different installation requirements. Each scenario's requirements are described in this document. Please choose only one scenario and implement its requirements.

Note: The North America Workshop events will only support the CloudLabs and Local PC (verification only) scenarios on the workshop day. The Local PC scenario is documented so the user may continue their learning after the workshop day is complete.

For a local PC installation, the files are large, so the internet speed needs to be fast.

2 CloudLab scenario

This scenario does not require any installation.

2.1 Account registration

ONE WARE AI: The lab user guide provides details on obtaining an account.

FPGA AI Suite requires access to Google Colab. This is where the AI model will be constructed and trained with a dataset. Follow the steps below to obtain a Google account. You may use an existing account, if available.

Click on this link to register a new Google account. Click Create account.

2.2 Hardware requirements

- Personal computer or laptop running Windows 11 with at least an Intel i5 core (or equivalent), 8GB RAM.

2.3 Software requirements

- Web browser.

3 Local PC (verification only) scenario

Two AI labs do require obtaining account credentials for online tool sets. Only follow the instructions below if you have chosen to do that lab.

3.1 Account registration

ONE WARE AI: The lab user guide provides details on obtaining an account.

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3.2 Hardware requirements

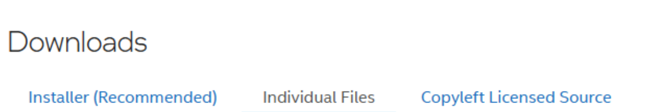
- Personal computer or laptop running Windows 10/11 with at least an Intel i5 core (or equivalent), 8GB RAM, 10GB hard drive space.
- AXC3000 Board
- USB C cable

3.3 Software requirements

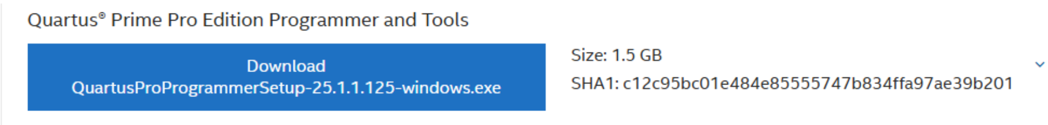
- Web browser
- Quartus Prime 25.1.1 Pro Standalone Programmer.
- Terminal emulator (Putty or Tera term)
- Python 3.13

3.3.1 Install Quartus Prime Pro Edition Programmer and Tools (25.1.1). Navigate to the Quartus Prime Pro 25.1.1 install page. Click on this link.

3.3.1.1 Scroll down and select the Individual Files tab under the Downloads section.

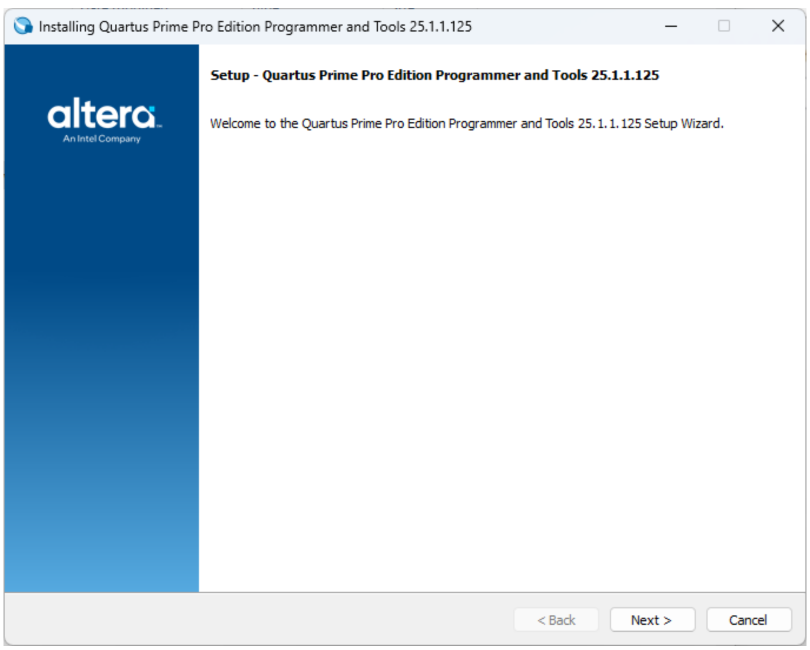


3.3.1.2 Scroll further to the Add-On and Stand-Alone Software section. Click on **Quartus Prime Pro Edition Programmer and Tools** to begin the download.



3.3.1.3 Click on the downloaded file to begin the installation. The file will typically be located in the Downloads folder by default.

3.3.1.4 Follow the installation instructions.



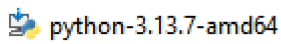
3.3.2 Install a terminal emulator.

- Install TeraTerm or
- Install Putty

3.3.3 Install Python

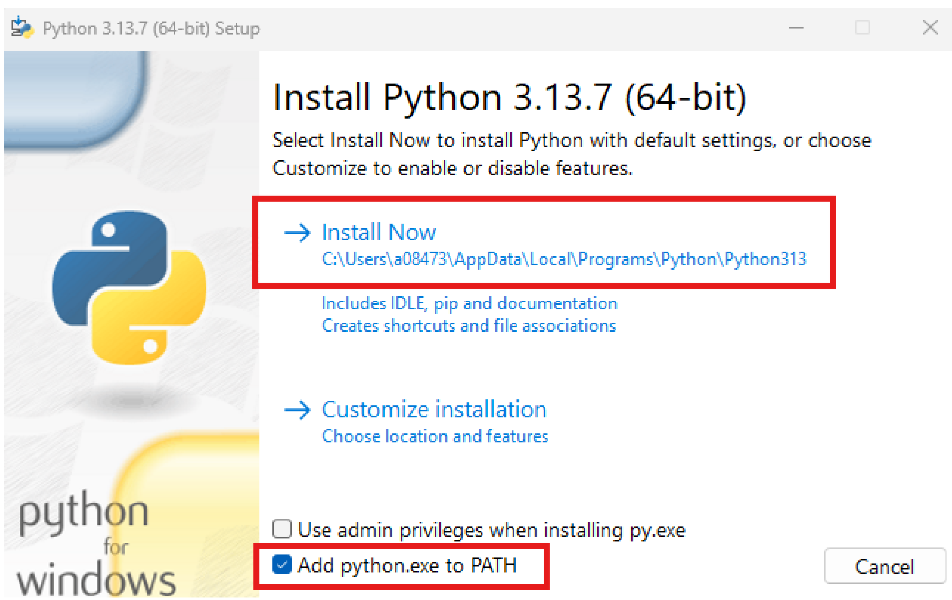
3.3.3.1 Download the installer

3.3.3.2 Right click the **installer** and select open.



3.3.3.3 Check the box ' **Add python.exe to PATH**'.

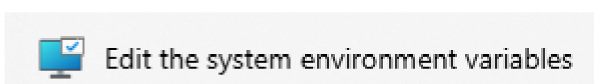
3.3.3.4 Select **Install Now**



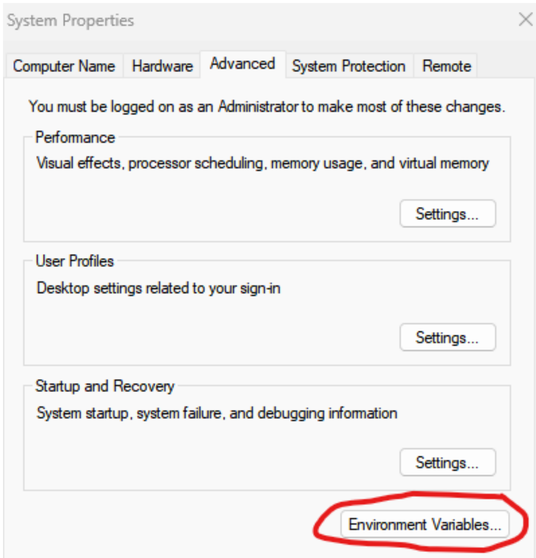
3.3.4 Add Environment Variable for System Console

3.3.4.1 Type Environment in the Windows search bar.

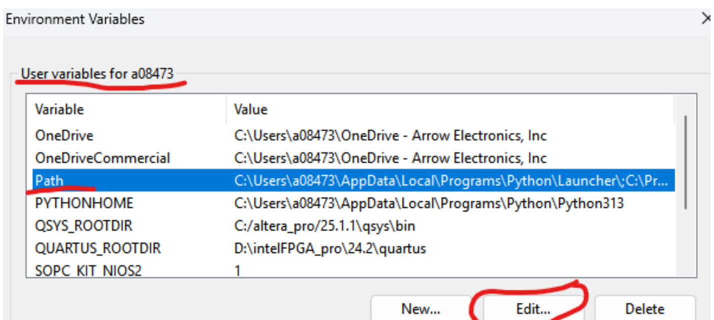
3.3.4.2 Select Edit **the system environment variables**



3.3.4.3 Press the Environment Variables button



3.3.4.4 Select PATH variable. Press Edit.



3.3.4.5 Press New

3.3.4.6 Add the following PATH

c:\altera_pro\25.1.1\qprogrammer\syscon\bin

3.3.5 Install Python Installer pip

3.3.5.1 Open a CMD shell and execute the following commands. If the ‘python’ command is not recognized, replace it with ‘py’.

curl https://bootstrap.pypa.io/get-pip.py -o get-pip.py

python get-pip.py

3.3.5.2 Install Python libraries. Execute the following in the CMD shell.

pip install numpy

pip install willow

pip install matplotlib

pip install telnetlib3

pip install serial

pip install pyserial

3.3.6 Install git for Windows

3.3.6.1 Use the Windows search bar to find **PowerShell**. Open PowerShell.

3.3.6.2 Copy the following **line of text** into the PowerShell and press enter.

winget install --id Git.Git -e --source winget

3.3.7 Installing USB-Blaster III on native Linux machines.

The Altera tools in general support Linux, but given the variety of distributions you may experience some challenges with programming cable(USB Blaster) drivers.

The following description has been verified on:

- Ubuntu 24.03.3 fresh install with default settings
- Quartus 25.3 fresh install into default location under home directory

```
# Step 1: Unplug USB Blaster cable and run these commands:
sudo tee /etc/udev/rules.d/92-usbblaster.rules >/dev/null <<'EOF'
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6001", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6002", MODE="0666"
```

```
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6003", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6010", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6810", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6020", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6022", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6024", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6025", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6026", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602C", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602D", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602E", MODE="0666"
EOF

sudo udevadm control --reload-rules
sudo udevadm trigger

# Step 2: Replug the USB Blaster cable

# Step 3 (Optional)
#   Make sure <altera_install_dir>/quartus/bin is in the PATH to find jtagd and
   jtagconfig executables...
pkill jtagd
jtagd --user-start
jtagconfig
```

4 Local PC scenario

Two AI labs do require obtaining account credentials for online tool sets. Only follow the instructions below if you have chosen to do that lab.

4.1 Account registration

ONE WARE AI: The lab user guide provides details on obtaining an account.

FPGA AI Suite requires access to Google Colab. This is where the AI model will be constructed and trained with a dataset. Follow the steps below to obtain a Google account. You may use an existing account, if available.

Click on this link to register a new Google account. Click Create account.

4.2 Hardware requirements

- Personal computer or laptop running Windows 10/11 with at least an Intel i7 core (or equivalent), 32GB RAM, 250GB hard drive space or more.
- AXC3000 Board
- USB C cable

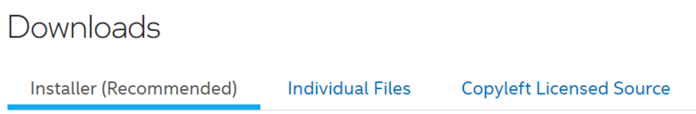
4.3 Software requirements

- Quartus Prime 25.1.1 Pro.
- WSL2 Ubuntu 22.04 virtual machine
- FPGA AI Suite
- ONE WARE Studio
- Terminal emulator (Putty or Tera term)

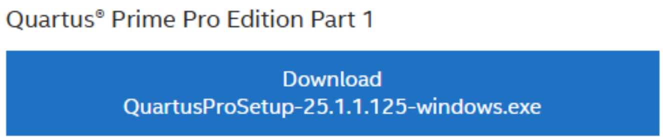
4.3.1 Quartus Prime 25.1.1 Pro.

4.3.1.1 Download Quartus Prime Pro 25.1.1 at the following link.

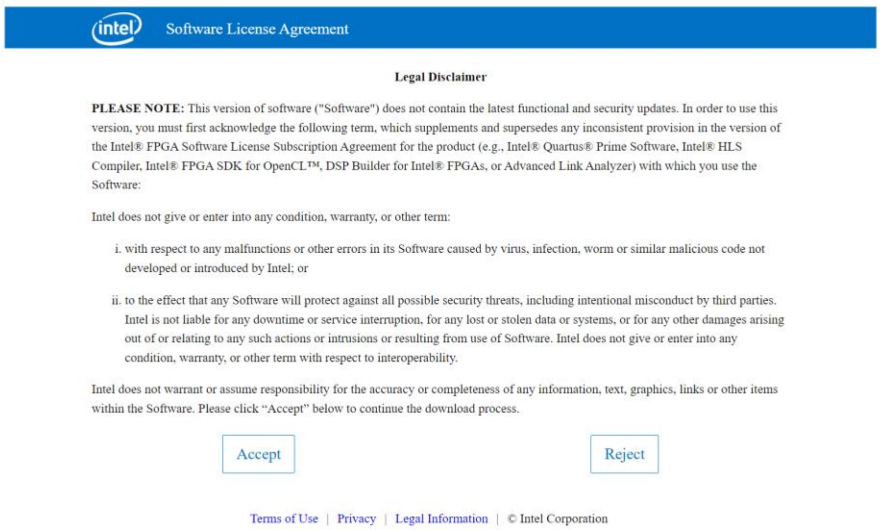
4.3.1.2 **Select the Individual Files** to reduce the amount of space needed on a computer. (The complete install could be downloaded, if wished but it will use more space.)



4.3.1.3 Select and download the Quartus Prime Pro Edition Part 1



4.3.1.4 When the Legal Disclaimer appears, read it and select **Accept**



Software License Agreement

Legal Disclaimer

PLEASE NOTE: This version of software ("Software") does not contain the latest functional and security updates. In order to use this version, you must first acknowledge the following term, which supplements and supersedes any inconsistent provision in the version of the Intel® FPGA Software License Subscription Agreement for the product (e.g., Intel® Quartus® Prime Software, Intel® HLS Compiler, Intel® FPGA SDK for OpenCL™, DSP Builder for Intel® FPGAs, or Advanced Link Analyzer) with which you use the Software:

Intel does not give or enter into any condition, warranty, or other term:

i. with respect to any malfunctions or other errors in its Software caused by virus, infection, worm or similar malicious code not developed or introduced by Intel; or

ii. to the effect that any Software will protect against all possible security threats, including intentional misconduct by third parties. Intel is not liable for any downtime or service interruption, for any lost or stolen data or systems, or for any other damages arising out of or relating to any such actions or intrusions or resulting from use of Software. Intel does not give or enter into any condition, warranty, or other term with respect to interoperability.

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Accept Reject

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4.3.1.5 It should start downloading.

4.3.1.6 **Select** the back button in the Browser to go back to the other files

4.3.1.7 Select the following files for download under the Individual files tab in the Individual files section.

Quartus® Prime Pro Edition Part 1

Download
QuartusProSetup-25.1.1.125-windows.exe

Quartus® Prime Pro Edition Part 2

Download
QuartusProSetup-part2-25.1.1.125-windows.qdz

Questa*-Intel FPGA and Starter Editions

Download
QuestaSetup-25.1.1.125-windows.exe

4.3.1.8 Select the following files for download under the Individual files tab in the Add-On and Stand-Alone Software files section.

Ashling* RiscFree* IDE for Altera

Download
RiscFreeSetup-25.1.1.125-windows.exe

Power Thermal Calculator

Download
ptc-25.1.1.125-windows.exe

4.3.1.9 Select the following files for download under the Individual files tab in the Devices files section.

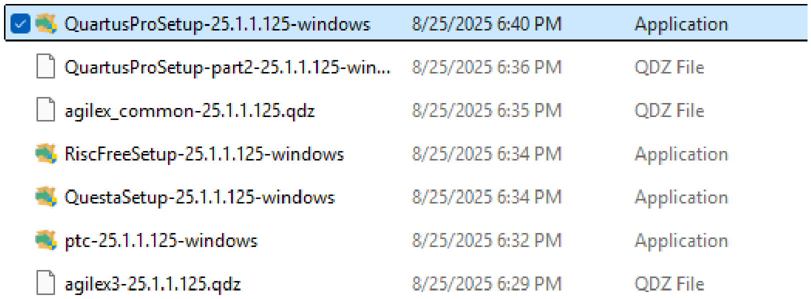
Agilex™ 3 device support (Requires Agilex™ common files)

Download
agilex3-25.1.1.125.qdz

Agilex™ common files

Download
agilex_common-25.1.1.125.qdz

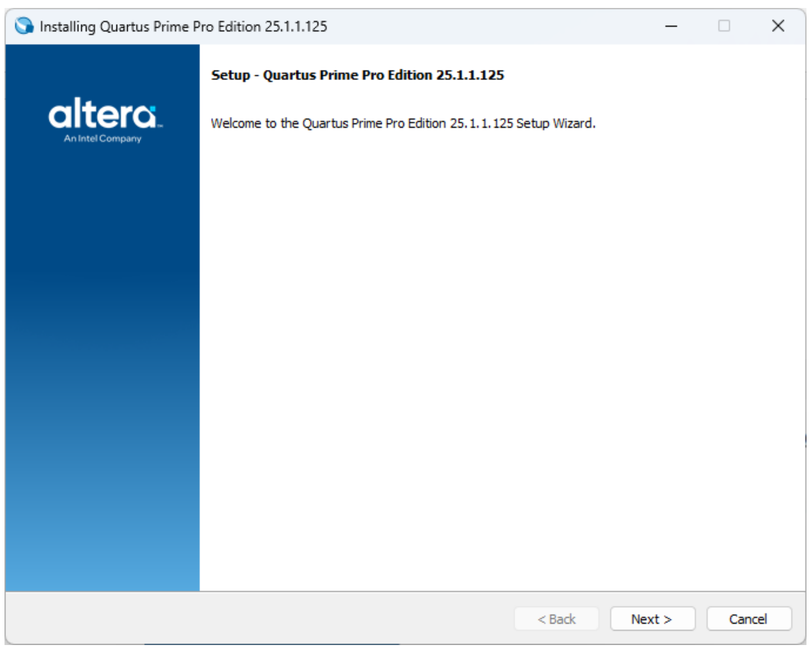
4.3.1.10 The downloaded files when viewed in Windows Explorer.



4.3.1.11 Select QuartusProSetup-25.1.1.125-windows. Right click, select open.

4.3.1.12 On some computers, it may be possible that you need to sign in for admin access in order to install software. It may be a while for the Quartus to appear.

4.3.1.13 The Quartus setup window will appear.



4.3.1.14 Select **Next**.

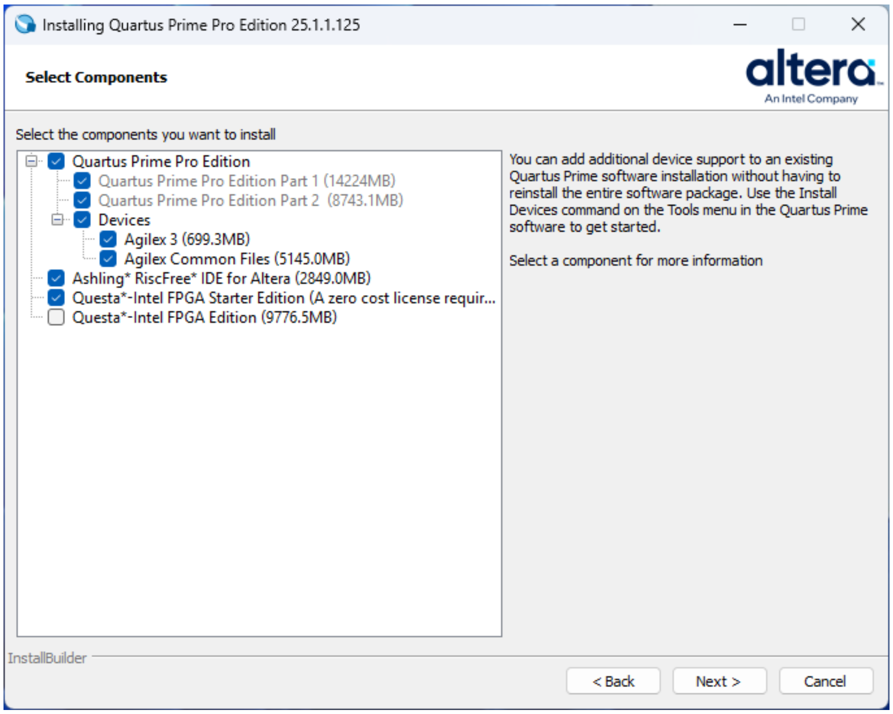
4.3.1.15 Read the License Agreement and select **I accept the agreement**.

4.3.1.16 Select **Next.**

4.3.1.17 Select the **default** directory.

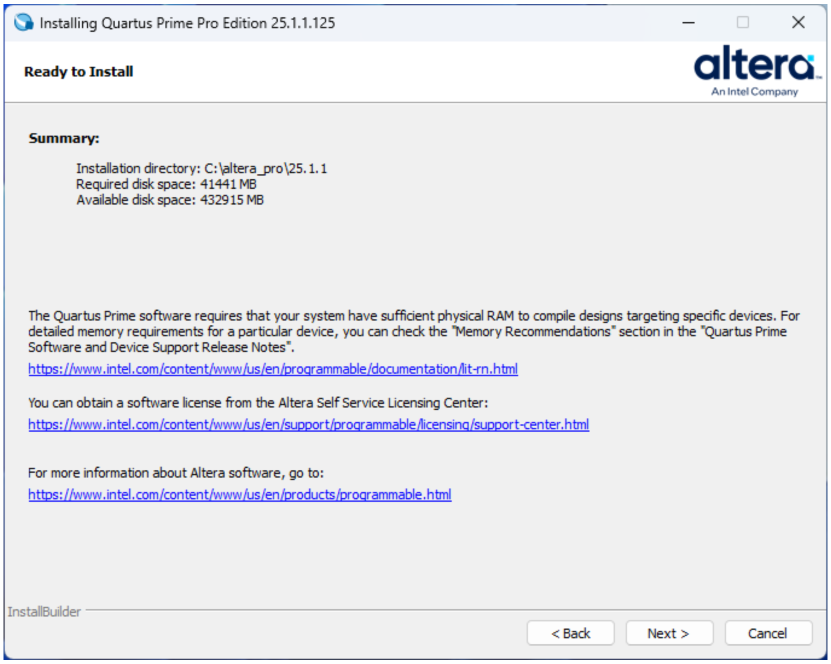
4.3.1.18 Select **Next**

4.3.1.19 Ensure that the following items are enabled

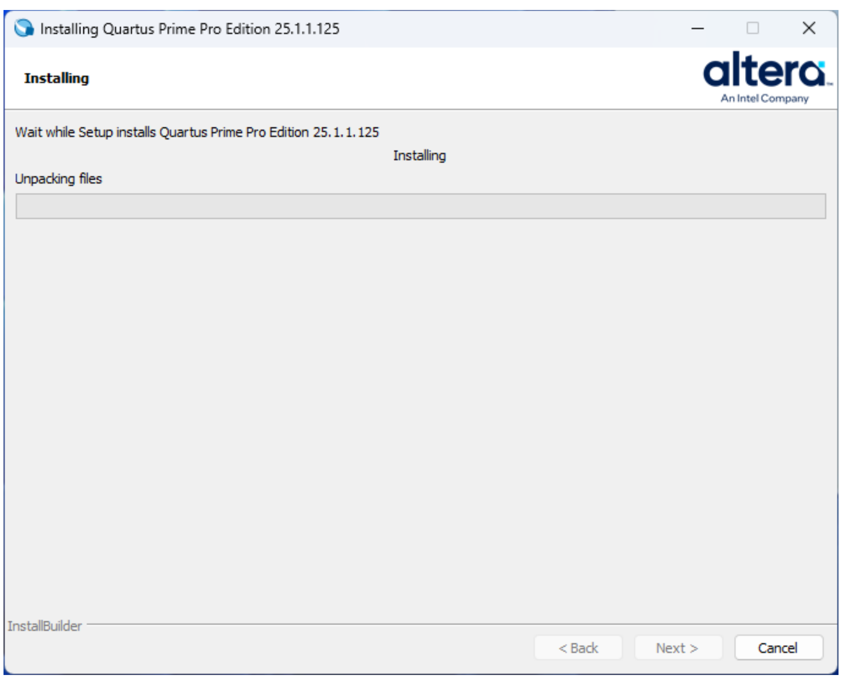


4.3.1.20 Select **Next**

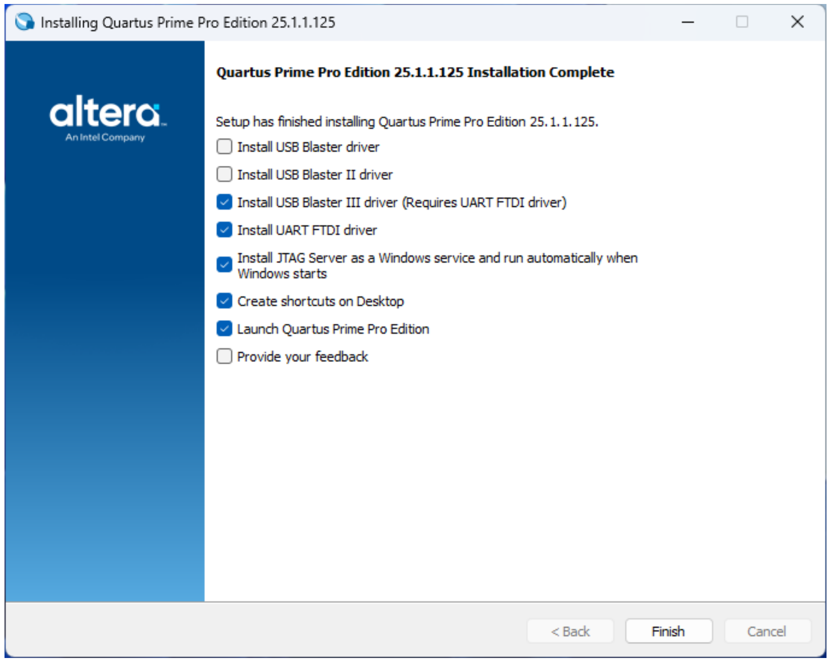
4.3.1.21 The Summary window will appear.



4.3.1.22 The install can take a while.



4.3.1.23 The setup installer will indicate when it is complete. A driver installation window will be shown. Uncheck the first two boxes.



4.3.1.24 Select Finish. Click through the driver install windows until complete.

4.3.2 Install WSL2 Ubuntu 22-04

4.3.2.1 Open Windows PowerShell with administrator privileges. Execute the following instructions to install Ubuntu 22.04 LTS

- dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart
- dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
- wsl --set-default-version 2
- reboot windows and open PowerShell with admin. priviledges
- wsl --install -d Ubuntu-22.04
    - select username = **altera**, password = **agilex3**, when prompted

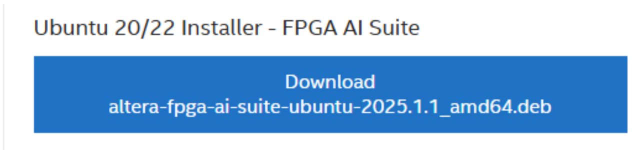
4.3.2.2 Invoke WSL2 with Ubuntu 22.04 LTS. Type Ubuntu in the Windows search bar. Click on the Ubuntu 22.04 LTS icon. Execute the following instructions in the Ubuntu shell.

- sudo apt update
- sudo apt install wget
- wget https://apt.kitware.com/ubuntu/pool/main/c/cmake/cmake_3.31.6-0kitware1ubuntu22.04.1_amd64.deb
- wget https://apt.kitware.com/ubuntu/pool/main/c/cmake/cmake-data_3.31.6-0kitware1ubuntu22.04.1_all.deb

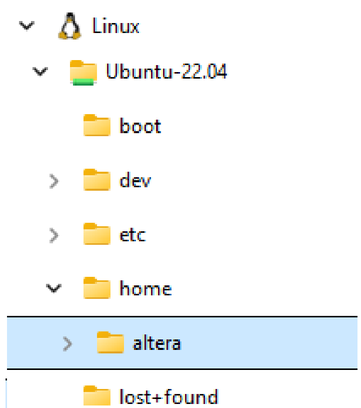
- sudo dpkg -i cmake-data_3.31.6-0kitware1ubuntu22.04.1_all.deb
- sudo dpkg -i cmake_3.31.6-0kitware1ubuntu22.04.1_amd64.deb

4.3.3 Install Altera FPGA AI Suite

4.3.3.1 Download Altera FPGA AI Suite. Click on this link to navigate to the download page. Click on Ubuntu 20/22 Installer - FPGA AI Suite.



4.3.3.2 Drop and drag the installation file from Windows into the Ubuntu filesystem. Open explorer in windows and navigate to the Linux icon. Expand and select the /home/altera folder. Drop the installation file into the /home/altera folder.



4.3.3.3 Copy the installation file to /opt/altera. If prompted, the password is **agilex3**. Execute the following instructions in the Ubuntu shell.

- sudo mkdir /opt/altera
- cd /opt/altera
- sudo cp ~/altera-fpga-ai-suite-ubuntu-2025.1.1_amd64.deb /opt/altera

4.3.3.4 Install FPGA AI Suite. Execute the following instructions in the Ubuntu shell.

- sudo apt install --reinstall libappstream4
- sudo apt install /opt/altera/altera-fpga-ai-suite-ubuntu-2025.1.1_amd64.deb
- if a dependency **error or warning** occurs do the following,   sudo apt --fix-broken install

4.3.4 Install the OpenVINO toolkit

4.3.4.1 Install required OpenVINO components. Upgrade to Python 3.12. Execute the following instructions.

- sudo apt update && sudo apt upgrade -y

- sudo add-apt-repository ppa:deadsnakes/ppa -y
- sudo apt update
- sudo apt install python3.12 python3.12-venv -y
- python3.12 -V

4.3.4.2 Create a new Python virtual environment. Execute the following instructions.

- mkdir ~/build-openvino-dev && cd ~/build-openvino-dev
- python3.12 -m venv openvino_env
- source openvino_env/bin/activate

4.3.4.3 Install the OpenVino 2024.6 Runtime. Execute the following instructions.

- sudo mkdir /opt/intel
- mkdir ~/Downloads && cd ~/Downloads

4.3.4.4 Download the OpenVINO 2024.6 Runtime archive file. Execute the following instructions.

```
curl -L https://storage.openvinotoolkit.org/\
repositories/openvino/packages/2024.6/linux/\
l_openvino_toolkit_ubuntu20_2024.6.0.17404.4c0f47d2335_x86_64.tgz \
--output openvino_2024.6.0.tgz
```

- tar -xf openvino_2024.6.0.tgz
- sudo mv l_openvino_toolkit_ubuntu20_2024.6.0.17404.4c0f47d2335_x86_64 /opt/intel/openvino_2024.6.0

4.3.4.5 Install system dependencies required by OpenVINO runtime.

- cd /opt/intel/openvino_2024.6.0/install_dependencies/
- sudo -E ./install_openvino_dependencies.sh

4.3.4.6 Create a symbolic link.

- cd /opt/intel
- sudo ln -s openvino_2024.6.0 openvino_2024

4.3.4.7 Configure openvino runtime environment variables.

- source /opt/intel/openvino_2024/setupvars.sh

4.3.4.8 Install openvino development tools.

- python3 -m pip install --upgrade pip
- pip install "openvino-dev[caffe, pytorch, tensorflow]==2024.6.0"

4.3.4.9 Install OpenCV.

- sudo apt update
- sudo apt install libopencv-dev python3-opencv

- pip install opencv-python (ignore numpy error??)

4.3.4.10 Configure OpenVINO environment.

- source ~/build-openvino-dev/openvino_env/bin/activate
- unset python_version
- source /opt/intel/openvino_2024/setupvars.sh

4.3.4.11 Add pytorch libraries.

- sudo apt update
- sudo apt install python3-pip -y
- pip3 install torch onnx torchsummary

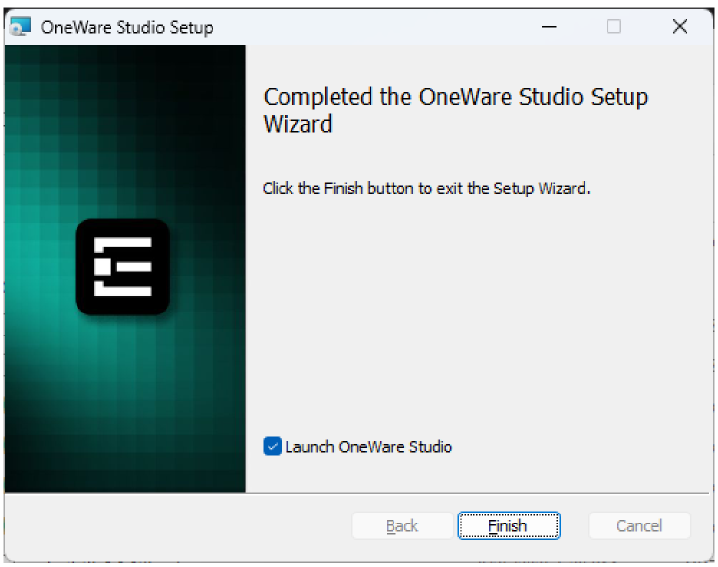
4.3.5 ONE WARE Studio

Only install if you will be completing the ONE WARE AI lab.

4.3.5.1 Click on this link to download the installer.

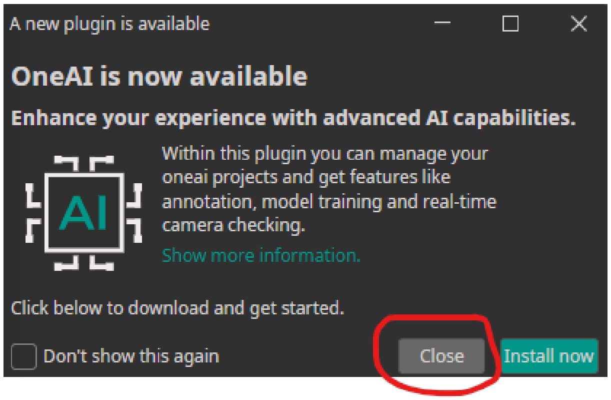
4.3.5.2 Then right click the file and select open.

4.3.5.3 Accept the license agreement and click through the installer. Click Finish.

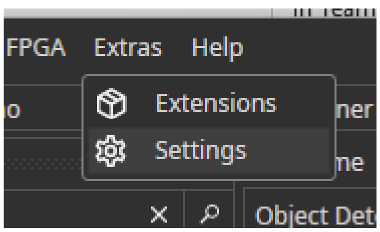


4.3.5.4 ONE WARE Studio will automatically launch. Complete the setup with the next steps.

4.3.5.5 Select Close when the OneAI plugin is recommended.

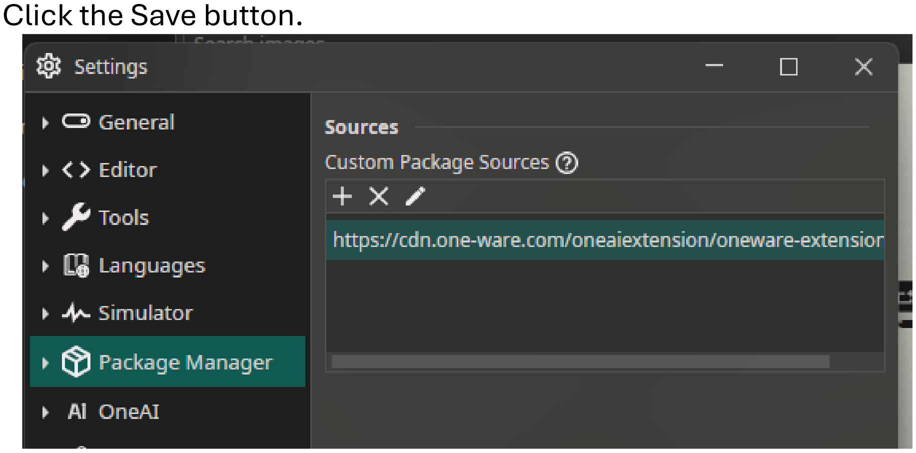


4.3.5.6 Select the Extras → Settings menu.

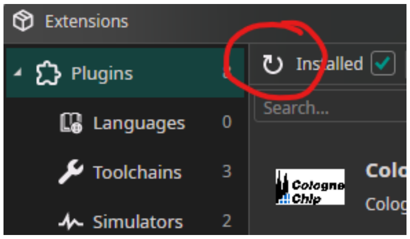


4.3.5.7 Go to the Package Manager tab. Add this path

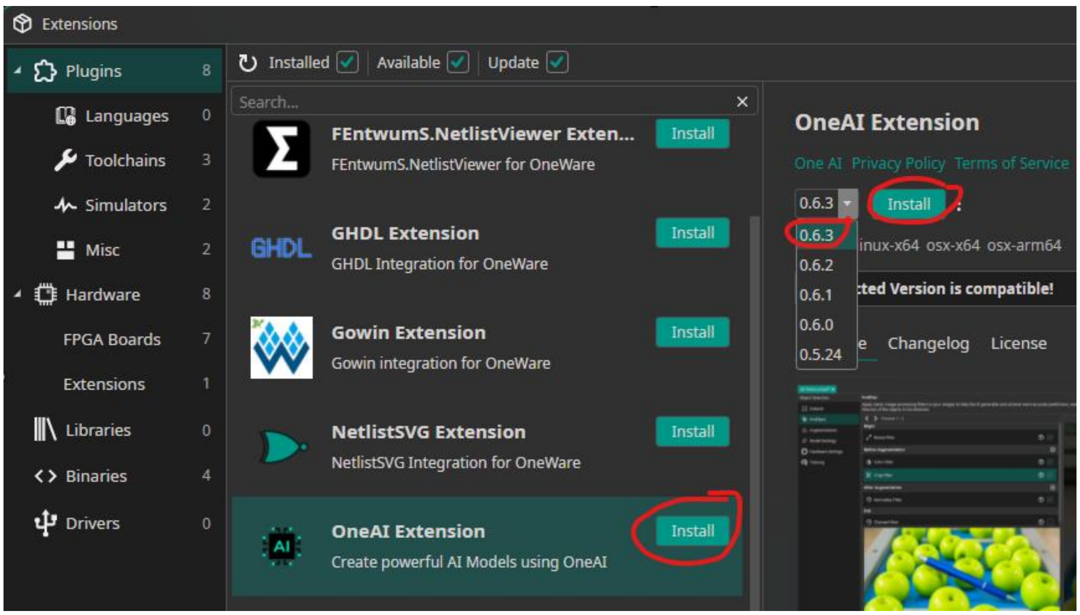
- https://cdn.one-ware.com/oneaiextension/oneware-extension.json
  Click the Save button.



4.3.5.8 Select the Extras → Extensions menu. Click on the refresh button.



4.3.5.9 Install The ONE AI Extension. Click Install. Select version 0.6.3 from the pulldown menu. Click Install.



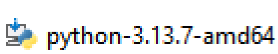
4.3.6 Install a terminal emulator.

- Install TeraTerm or
- Install Putty

4.3.7 Install Python

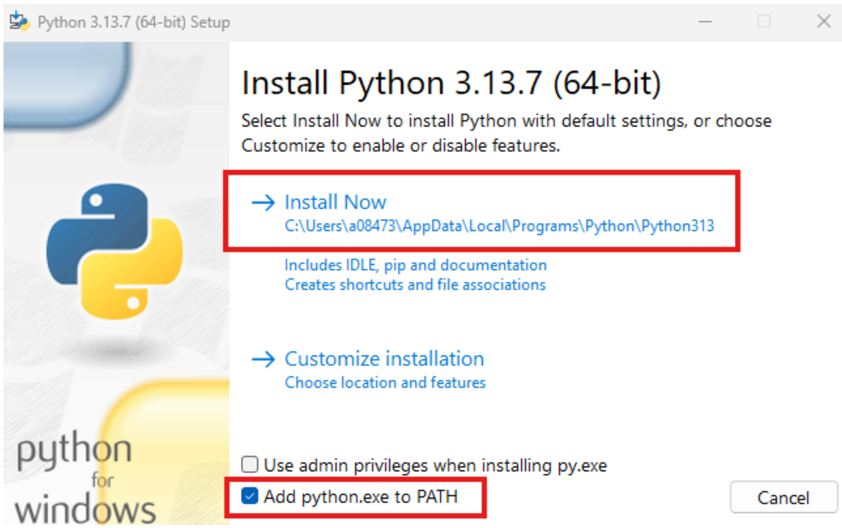
4.3.7.1 Download the installer

4.3.7.2 Right click the **installer** and select open.



4.3.7.3 Check the box ' **Add python.exe to PATH**'.

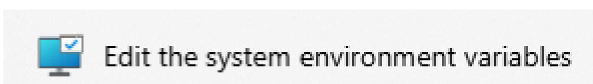
4.3.7.4 Select **Install Now**



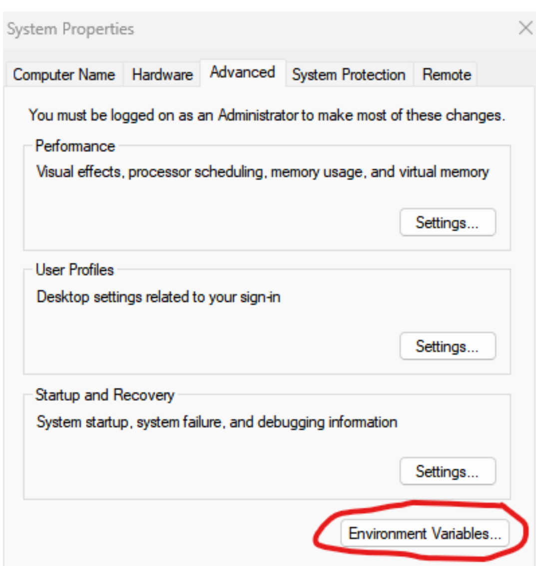
4.3.8 Add Environment Variable for System Console

4.3.8.1 Type Environment in the Windows search bar.

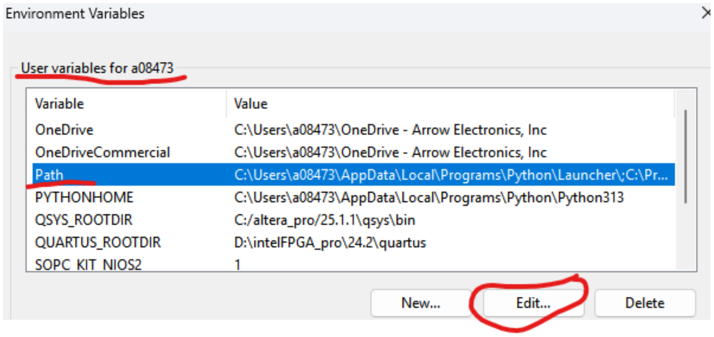
4.3.8.2 Select Edit **the system environment variables**



4.3.8.3 Press the Environment Variables button



4.3.8.4 Select PATH variable. Press Edit.



4.3.8.5 Press New

4.3.8.6 Add the following PATH

c:\altera_pro\25.1.1\syscon\bin

4.3.9 Install Python Installer pip

4.3.9.1 Open a CMD shell and execute the following commands. If the 'python' command is not recognized, replace it with 'py'.

curl https://bootstrap.pypa.io/get-pip.py -o get-pip.py

python get-pip.py

4.3.9.2 Install Python libraries. Execute the following in the CMD shell.

pip install numpy

pip install willow

pip install matplotlib

pip install telnetlib3

pip install serial

pip install pyserial

4.3.10 Install git for Windows

4.3.10.1 Use the Windows search bar to find **PowerShell**. Open PowerShell.

4.3.10.2 Copy the following **line of text** into the PowerShell and press enter.

winget install --id Git.Git -e --source winget

4.3.11 Installing USB-Blaster III on native Linux machines.

The Altera tools in general support Linux, but given the variety of distributions you may experience some challenges with programming cable(USB Blaster) drivers.

The following description has been verified on:

- Ubuntu 24.03.3 fresh install with default settings
- Quartus 25.3 fresh install into default location under home directory

```
# Step 1: Unplug USB Blaster cable and run these commands:
sudo tee /etc/udev/rules.d/92-usbblaster.rules >/dev/null <<'EOF'
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6001", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6002", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6003", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6010", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6810", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6020", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6022", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6024", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6025", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="6026", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602C", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602D", MODE="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="09fb", ATTRS{idProduct}=="602E", MODE="0666"
EOF

sudo udevadm control --reload-rules
sudo udevadm trigger

# Step 2: Replug the USB Blaster cable

# Step 3 (Optional)
#   Make sure <altera_install_dir>/quartus/bin is in the PATH to find jtagd and
   jtagconfig executables...
pkill jtagd
jtagd --user-start
jtagconfig
```

5 Licensing

Quartus Prime Pro requires a free license for Agilex 3 devices. Licenses are also required for the Nios V/g and MIPI CSI-2 IP cores. The Questa are requires a license.

5.1 Identify the PC NIC ID

5.1.1 Open a Windows PowerShell

5.1.2 Type ipconfig /all. The NIC is the Physical Address of the adapter. This will be used to create a new computer for licensing purposes.

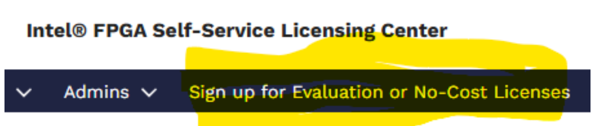
```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Ethernet Connection (22) I219-LM
   Physical Address. . . . . . . . . : AC-B4-80-94-58-7B
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

5.2 Create Evaluation Licenses

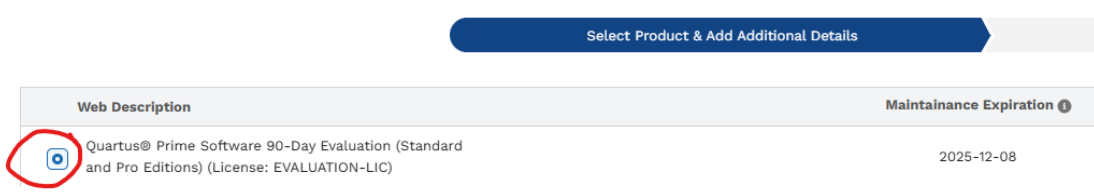
5.2.1 Navigate to the Self Service Licensing Center

5.2.2 Sign in to an existing account or Enroll to get a new account.

5.2.3 Sign up for Evaluation or No-Cost Licenses

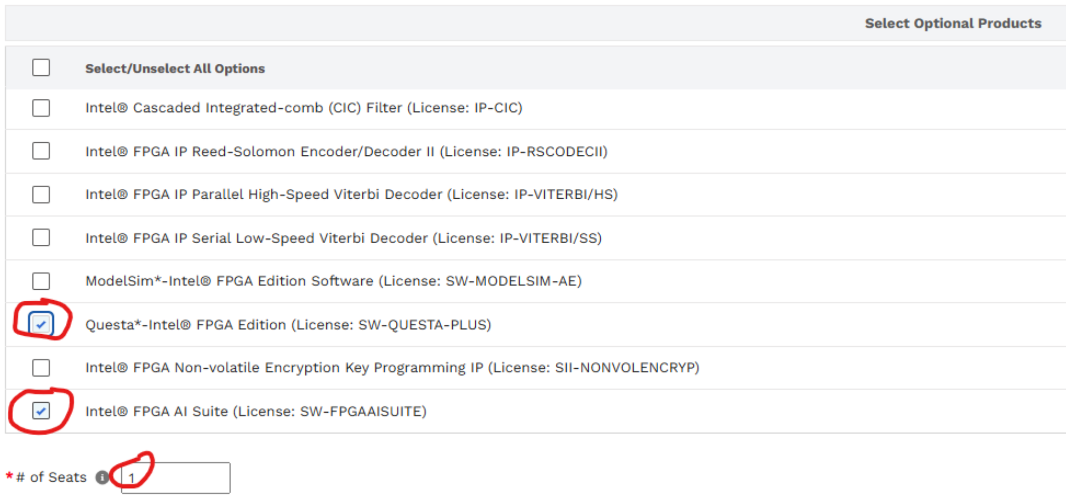


5.2.4 Select Quartus Prime Software 90-Day Evaluation License



5.2.4.1 Scroll to the bottom of the page

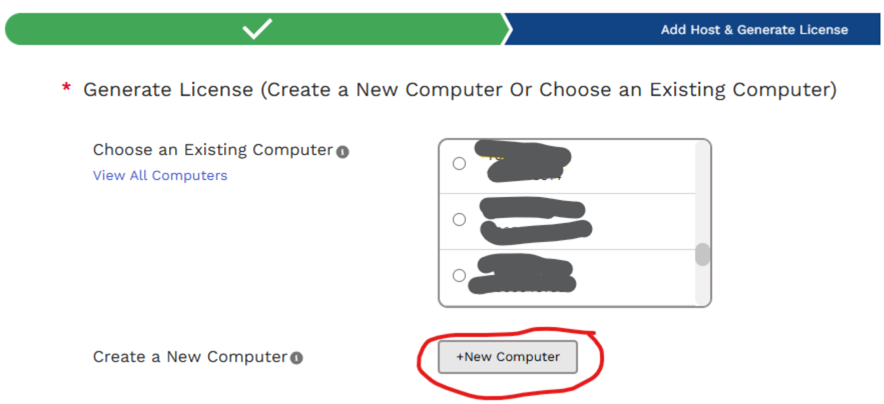
5.2.4.2 Select Intel FPGA AI Suite and Questa-Intel FPGA Edition



5.2.4.3 Click Next

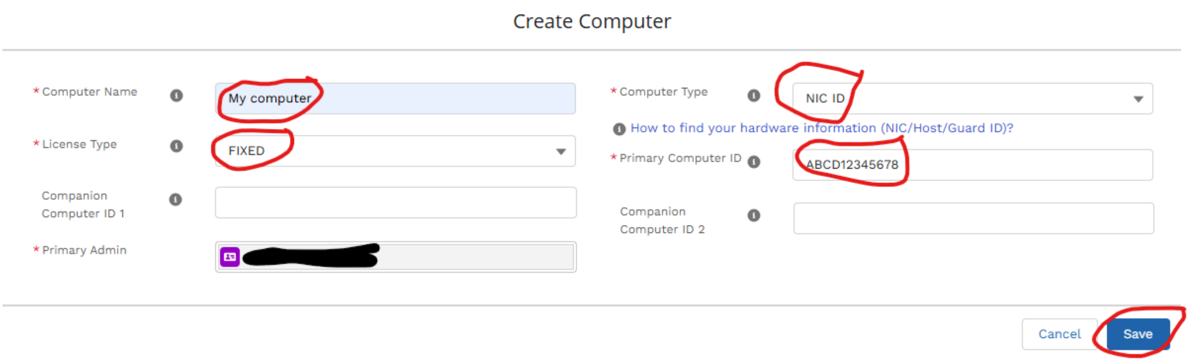
5.2.5 Add a New Computer

5.2.5.1 Click on +New Computer



5.2.5.2 Create Computer

- Computer Name: Enter a unique name
- License Type: Fixed
- Computer Type: NIC ID
- Primary Computer: Enter 12 digit HEX value of the computer NIC ID



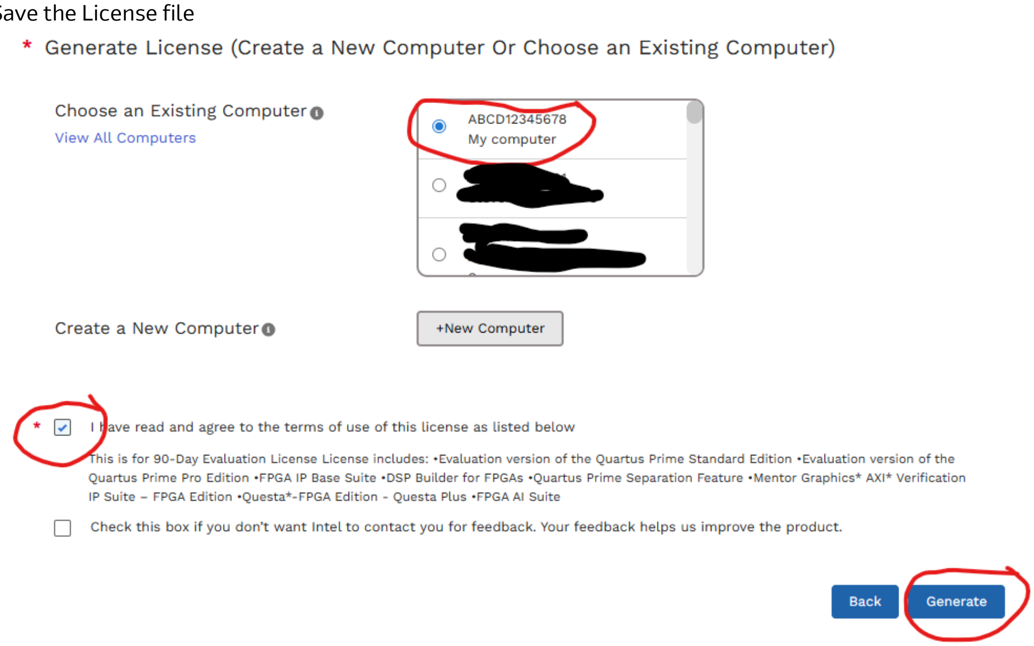
5.2.5.3 Press Save

5.2.5.4 Select the created computer

5.2.5.5 Click the License check box

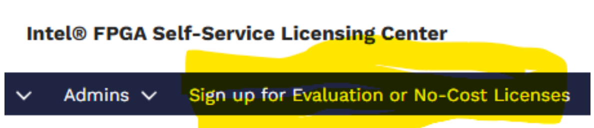
5.2.5.6 Press Generate

Save the License file



5.2.6 Add a Nios V/m IP license

5.2.6.1 Select **Sign up for Evaluation or No-Cost Licenses** again



5.2.6.2 Select the Nios V/m Microcontroller FPGA IP

5.2.6.3 Click **Next**

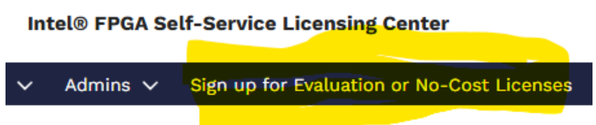
5.2.6.4 Select the **existing computer** that was created in 5.2.5.2

5.2.6.5 Click the License check box

5.2.6.6 Press Generate

5.2.7 Add a MIPI DSI/CSI-2 FPGA IP license

5.2.7.1 Select **Sign up for Evaluation or No-Cost Licenses** again



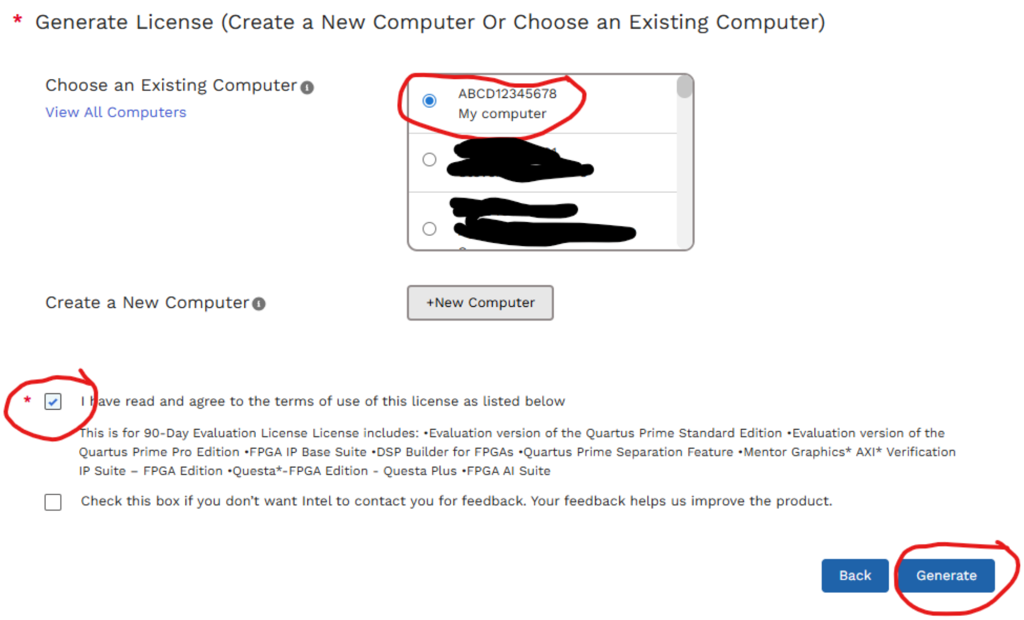
5.2.7.2 Select the MIPI DSI/CSI-2 FPGA IP

5.2.7.3 Click **Next**

5.2.7.4 Select the **existing computer** that was created in 5.2.5.2

5.2.7.5 Click the License check box

5.2.7.6 Press Generate



5.2.8 Save the License files

A total of five emails and licenses will be sent to the email address on record.

5.2.8.1 Save the **last two** licenses sent. The remainder are **duplicates**. A good location could be **c:\altera_pro**

5.2.9 Merge the two license files into one

5.2.9.1 Open the **two license** files in a text editor (Notepad or Notepad++ for example)

5.2.9.2 **Cut and paste** the contents of the first license file into the second license file.

5.2.9.3 Make a note of the second license **filename** and save it.

5.2.10 Add Environment variables for Quartus and Questa

5.2.10.1 Open the System Properties window by pressing the **Windows key** and "R". This will open the Run window.

5.2.10.2 Type the following Run command: **SystemPropertiesAdvanced** and press **enter**

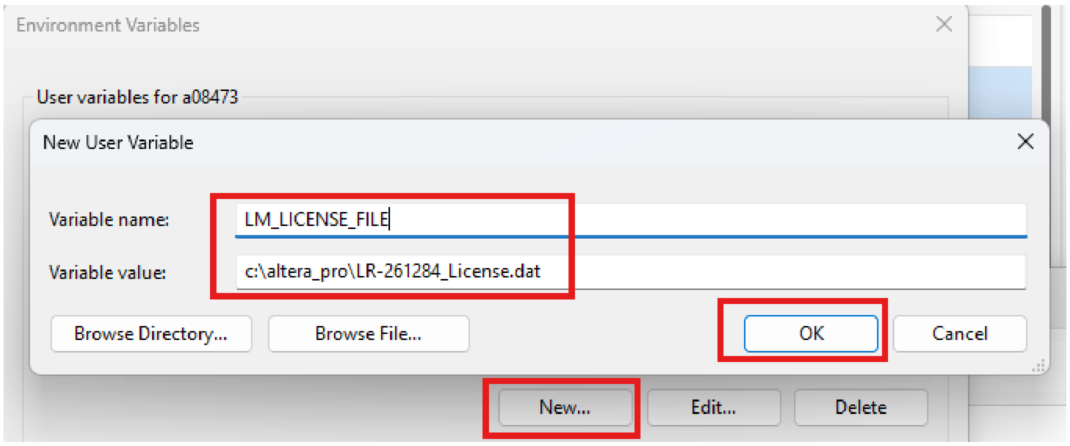
5.2.10.3 Click the Environment Variables **button**

5.2.10.4 Press **New** to create a new User Environment Variable

5.2.10.5 Enter **LM_LICENSE_FILE** for the **Variable name**

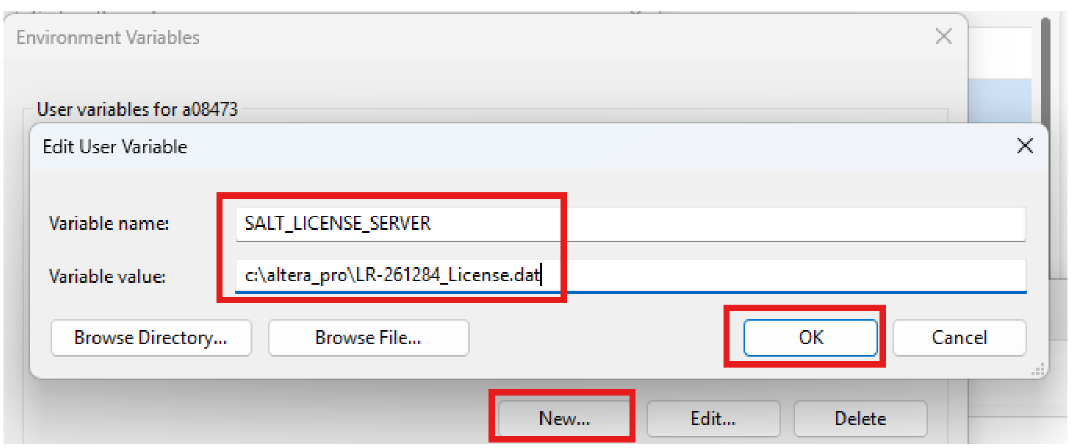
5.2.10.6 Enter the path to the license file for the **Variable value**

5.2.10.7 Press OK



5.2.10.8 Press **New** to add a new environment variable.

5.2.10.9 Repeat the steps above to add the **Questa** environment variable



5.2.10.10 Press OK twice to exit System Properties

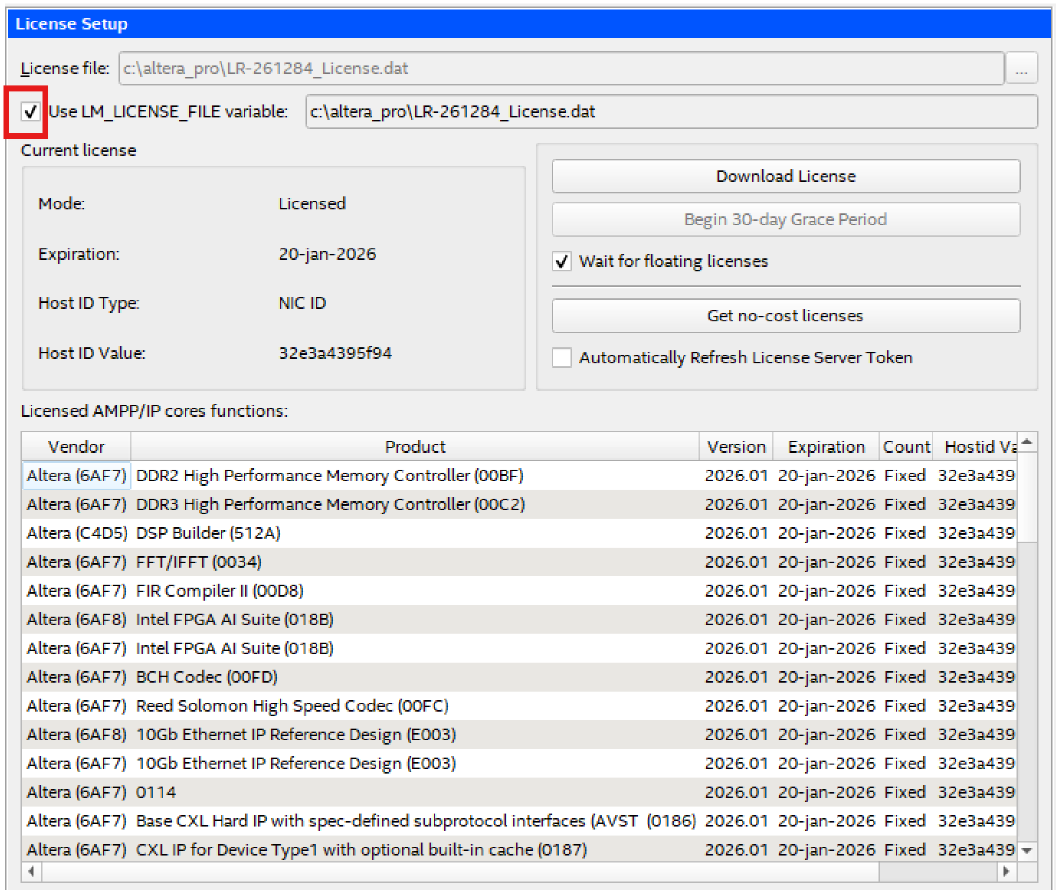
5.2.11 Set up the Quartus License

5.2.11.1 Open Quartus

5.2.11.2 Open the Quartus License setup page. Tools → License Setup

5.2.11.3 Navigate to the **Use LM_LICENSE_FILE variable**. Click the **check box**

5.2.11.4 Press OK



5.2.12 Set the environment variable for Questa

5.2.12.1 Create a new environment variable named SALT_LICENSE_SERVER

5.2.12.2 Point to the License file on the computer.

6 Legal Disclaimer

ARROW ELECTRONICS

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